Alberta Surface Water Quality Dashboard - Summary Report by Kai Wong (May 19, 2026)

Github repository: https://github.com/kaionwong/surface-water-quality-dashboard

Deployed Streamlit app: https://surface-water-quality-dashboard-8kkvmcckcfq5rarcsrgxy4.streamlit.app/

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1. Key Insights

- Approximately 0.43% of all records were classified as unreliable, while 1.16% were classified as either unreliable or potentially unreliable.
  - "Unreliable" records included the following qualifier combinations: `SUS`, `RER|SUS`, `HT|RER|SUS`, `HT|SUS`, and `DR|SUS`.
  - "Potentially unreliable" records included: `HT`, `HT|RER`, `FSE|HT`, `DR|HT`, `DR`, `DR|FSE`, `DR|SPNF`, and `SPNF`.
- Clear spatial differences in data quality were observed across monitoring stations. For example, most stations in the Grande Prairie region had 0% unreliable records, whereas many stations across the rest of Alberta showed unreliable record rates between >0% and 5%.
- More detailed spatial and risk-related patterns emerged when analyzing specific combinations of variables (all variables or single variable) and stations (all stations or single station).
- Several variables exhibited recognizable seasonal behaviour alongside broader long-term trends:
  - Variables such as "Residue Filterable", "Macrophyte Cover", and "Hardness Total CaCO3" showed recurring seasonal patterns with relatively stable long-term trends.
  - Variables including "Ice Thickness", "Sum of Anions", and "Sum of Cations" showed seasonal patterns combined with increasing or decreasing long-term trends.
  - "Fluorescent Dissolved Organic Matter (FDOM) – Field" displayed a non-seasonal but increasing long-term trend.
- SUSPECT (`SUS`) data quality flags were most common among measurements such as:
  - Residue Filterable (7.79%)
  - Total Dissolved Solids (5.43%)
  - pH (Field) (4.10%)
  - pH (Lab) (2.51%)
- In relative terms, the stations with the highest percentage of unreliable records were:
  1. Tay River (downstream of pipeline crossing) - 2.78%
  2. Michichi Creek (near the mouth) - 2.10%
  3. Pipestone Creek (upstream of Millet STP Outfall) - 1.98%
- In absolute terms, the stations with the highest number of records flagged as SUSPECT were:
  1. North Saskatchewan River (at WSC Gauge at Whirlpool Point) - 33 records
  2. Peace River (upstream of Smoky River near Shaftesbury Ferry Transect) - 27 records
  3. North Saskatchewan River (at Saunders Campground – transect) - 25 records
  4. Peace River (1.5 km above the confluence with the Whitemud River – transect) - 25 records

2. Future Directions

- Large language models (LLMs) could be applied to the `SampleComment` field to extract recurring themes, identify possible causes of data quality issues, and group related types of anomalies or operational concerns.
- To test core quality assumptions, a direction is to build ML models that learn patterns from SUS (and other flagged) records using all other available fields, then apply these patterns to currently "good quality" (unflagged) data to identify records that may be SUS-like but were not originally flagged.